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The Editors, *Advanced Intelligent Systems*

Dear Editors,

We are pleased to submit our manuscript, “Optimal stimulation sites are not the most affected: personalised fMRI models of Alzheimer’s disease,” by Cristiano Capone, Enza Cece, Luca Falorsi, Andrea Ciardiello, Guido Gigante, Evaristo Cisbani and Maurizio Mattia, for consideration in *Advanced Intelligent Systems*.

Targeted neuromodulation rests on two decisions that are still made heuristically: which site to engage, and at what amplitude. Both are typically settled anatomically and fixed in advance. We ask whether they can instead be *computed*, for each individual, from a model of that person’s own brain.

We build individualised digital twins: reservoir-computing models fit to resting-state functional MRI so that their free-running dynamics reproduce each individual’s own lagged connectivity. The design choice that makes this work is that the recurrent substrate is fixed and shared across every subject, and only a linear read-out is fit per person. Three consequences follow. Per-subject estimation stays tractable where a subject-specific autoregression would be underdetermined, since 121 regions must be fit from roughly 140 volumes. All fitted models occupy the same coordinates, so they can be compared, averaged and interpolated. And the reservoir’s separate input pathway makes “drive region k ” an explicit, well-posed operation rather than an ad hoc addition. The resulting twins are identifiable: a model fit to one individual describes that individual’s held-out data better than anyone else’s.

Perturbing them yields a result with a direct engineering consequence. Correcting a twin’s connectivity toward a control template reverts the disease signature, but the necessary correction is distributed, and no single-node edit reproduces it. A focal drive at the region whose connectivity is most altered fails at any amplitude, whereas a drive at the region selected by its *effect on the disease discriminant* succeeds from one site. The two site sets are nearly disjoint, and the reason is mechanistic rather than incidental: the most-affected regions are the most weakly connected, and therefore poor actuators. Targets are individual, and a real-time closed-loop controller that titrates amplitude from the model’s own output reaches comparable efficacy at substantially lower dose, using only causally available information.

We are explicit about scope. This is an in-silico study; no patient was stimulated, and complete reversal requires amplitudes above the physiological range. The contribution is a computable, per-patient prescription for where and how hard to drive, and a demonstration that the intuitive target is systematically the wrong one. We believe this speaks to readers of *Advanced Intelligent Systems* working on digital twins, closed-loop control and model-based decision-making in biological systems, beyond the Alzheimer’s application in which we demonstrate it.

The manuscript is original, is not under consideration elsewhere, and all authors have approved its submission. The data derive from the Alzheimer’s Disease Neuroimaging Initiative and are used under its Data Use Agreement. Code is available at github.com/cristianocapone/AD-reservoir-FC-stimulation. We declare no competing interests, and have suggested potential referees who have no conflict of interest with the authors. We thank you for considering our work.

Sincerely,

Cristiano Capone, on behalf of all authors